

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

NND PROJECT REPORT



COURSE NAME : NETWORK & NETWORKING DEVICES

COURSE ID : INITC602

**PROJECT TITLE : Router and Switch Management with TFTP
Backup & IOS Upgrade**

SUBMITTED TO : Ms. Pooja Yadav

SUBMITTED BY : Vansh Bhardwaj (2023UIN3324)

Khushi Kumari (2023UIN3334)

Ishita Soni (2023UIN3343)

TABLE OF CONTENTS

S.No.	Detail
1.	Introduction
2.	Network Topology Design
3.	Cisco Packet Tracer Setup
4.	End Device IP Configuration
5.	Switch Configuration
6.	Router Configuration
7.	OSPF Verification & Connectivity Testing
8.	TFTP Configuration Backup
9.	IOS Image Backup & Upgrade via TFTP
10.	Advanced Configuration & Security
11.	Complete Verification
12.	Simulation Mode in Packet Tracer

1. Introduction

Modern enterprise networks rely heavily on centralized management to ensure operational continuity and swift disaster recovery. This project simulates an enterprise-grade university network using Cisco Packet Tracer 9.0. The topology models a multi-department university campus with centralized network management, TFTP-based configuration backup and restoration, and IOS image management. By deploying a Trivial File Transfer Protocol (TFTP) server, the project provides a centralized repository to store router and switch configurations (**running-config** and **startup-config**) and manage Internetwork Operating System (IOS) image upgrades. This ensures that in the event of hardware failure or accidental misconfiguration, the network devices can be quickly restored with minimal downtime.

1.1 Project Objectives

- Design & implement a hierarchical 3-tier campus network topology
- Configure inter-VLAN routing using Layer 3 switching and router-on-a-stick
- Deploy OSPF as the dynamic routing protocol across all routers
- Set up a reliable centralized TFTP server for configuration and IOS image management
- Automate configuration backup from all routers and switches to TFTP
- Simulate an IOS upgrade process via TFTP on selected devices
- Implement access control and security best practices (SSH, ACLs, port security)
- Validate full end-to-end connectivity between all VLANs and departments

1.2 Learning Outcomes

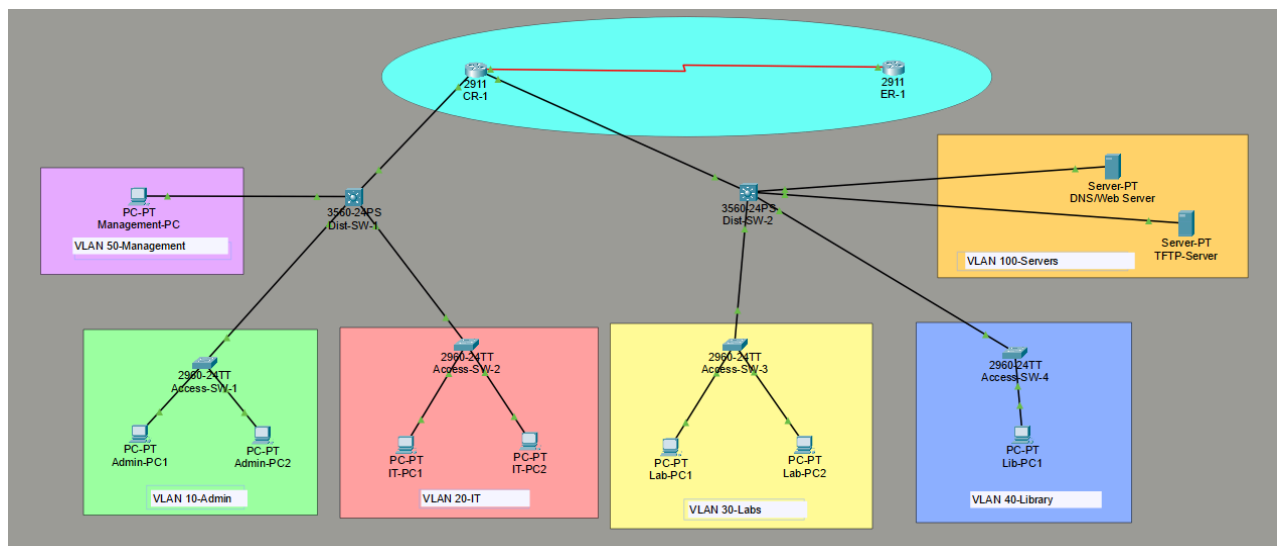
- Understand centralized network management concepts
- Master TFTP operations (copy, backup, restore) in Cisco IOS
- Practice enterprise VLAN design and trunking
- Gain hands-on experience with OSPF configuration and verification
- Learn IOS image management workflows

2. Network Topology Design

2.1 Topology Overview

The network follows a three-tier hierarchical model: Core, Distribution, and Access layers. This mirrors real-world enterprise and university campus deployments.

Layer	Device Name	Role
Core Layer	Core-Router (CR-1)	Central routing hub, OSPF ABR, gateway to ISP/Internet
Distribution	Dist-SW-1 (L3 Switch)	VLAN routing for Admin & IT blocks
Distribution	Dist-SW-2 (L3 Switch)	VLAN routing for Labs & Library
Access Layer	Access-SW-1	Admin Department end-user switch
Access Layer	Access-SW-2	IT Department end-user switch
Access Layer	Access-SW-3	Computer Lab switch
Access Layer	Access-SW-4	Library switch
WAN/Edge	Edge-Router (ER-1)	WAN edge, NAT, DHCP server
Management	TFTP-Server	Configuration & IOS backup server
Management	Management-PC	Network admin workstation
End Devices	Admin-PC-1, Admin-PC-2	Admin department hosts
End Devices	IT-PC-1, IT-PC-2	IT department hosts
End Devices	Lab-PC1, Lab-PC2	Computer lab hosts
End Devices	Lib-PC1	Library host
Services	DNS/Web Server	Internal services server



2.2 VLAN Design

VLANs provide logical segmentation between departments, improving security and broadcast control.

VLAN ID	Name	Subnet	Description
VLAN 10	Admin	192.168.10.0/24	Administrative Dept
VLAN 20	IT	192.168.20.0/24	IT Dept
VLAN 30	Labs	192.168.30.0/24	Computer Labs
VLAN 40	Library	192.168.40.0/24	Library Network
VLAN 50	Management	192.168.50.0/24	Network Management (TFTP, Admin PC)
VLAN 99	Native/Trunk	N/A	Native VLAN for trunk links
VLAN 100	Servers	192.168.100.0/24	Internal Servers (DNS, Web, TFTP)

2.3 IP Addressing Scheme

Device	Interface	IP Address	Notes
Core-Router (CR-1)	Gi0/0 (to Dist-SW-1)	10.0.1.1/30	Point-to-point link
Core-Router (CR-1)	Gi0/1 (to Dist-SW-2)	10.0.2.1/30	Point-to-point link
Core-Router (CR-1)	Se0/0/0 (to ER-1)	10.0.0.1/30	WAN serial link
Edge-Router (ER-1)	Se0/0/0 (to CR-1)	10.0.0.2/30	WAN serial link
Edge-Router (ER-1)	Gi0/0 (Internet-side)	DHCP/ Static ISP	Internet-facing interface
Dist-SW-1	VLAN 10 SVI	192.168.10.1/24	Admin gateway
Dist-SW-1	VLAN 20 SVI	192.168.20.1/24	IT gateway
Dist-SW-1	VLAN 50 SVI	192.168.50.1/24	Management gateway
Dist-SW-1	Uplink to CR-1	10.0.1.2/30	L3 routed link
Dist-SW-2	VLAN 30 SVI	192.168.30.1/24	Labs gateway
Dist-SW-2	VLAN 40 SVI	192.168.40.1/24	Library gateway
Dist-SW-2	VLAN 100 SVI	192.168.100.1/24	Servers gateway
Dist-SW-2	Uplink to CR-1	10.0.2.2/30	L3 routed link
TFTP-Server	NIC	192.168.100.10/24	GW: 192.168.100.1
Management-PC	NIC	192.168.50.10/24	GW: 192.168.50.1

Admin-PC1	NIC	192.168.10.10/24	GW: 192.168.10.1
Admin-PC2	NIC	192.168.10.11/24	GW: 192.168.10.1
IT-PC1	NIC	192.168.20.10/24	GW: 192.168.20.1
IT-PC2	NIC	192.168.20.11/24	GW: 192.168.20.1
Lab-PC1	NIC	192.168.30.10/24	GW: 192.168.30.1
Lab-PC2	NIC	192.168.30.11/24	GW: 192.168.30.1
Lib-PC1	NIC	192.168.40.10/24	GW: 192.168.40.1
DNS/Web Server	NIC	192.168.100.20/24	GW: 192.168.100.1

3. Cisco Packet Tracer Setup

3.1 Software Requirements

- Cisco Packet Tracer Version: 9.0 or higher
- Operating System: Windows 10/11, macOS 12+, or Ubuntu 20.04+
- RAM: Minimum 8 GB (16 GB recommended for 15+ devices)
- Disk: 5 GB free space

3.2 Device Selection in Packet Tracer

The following device categories are used in Cisco Packet Tracer :

Device	CPT Category / Model
Core-Router (CR-1)	Routers > 2911
Edge-Router (ER-1)	Routers > 2911
Dist-SW-1, Dist-SW-2	Switches > 3560-24PS
Access-SW-1 to SW-4	Switches > 2960-24TT
TFTP-Server, DNS/Web Server	End Devices > Server
Management-PC, Admin-PCs, IT-PCs, Lab PCs, Lib-PC	End Devices > PC

3.3 Physical Topology Connections

Connect devices using the following cable types. In CPT, select the correct cable from the Connections panel (lightning bolt icon at bottom).

From	To	Cable Type
CR-1 Gi0/0	Dist-SW-1 Gi0/1	Straight-through Copper
CR-1 Gi0/1	Dist-SW-2 Gi0/1	Straight-through Copper
CR-1 Se0/0/0	ER-1 Se0/0/0	Serial DCE/DTE (Serial)
Dist-SW-1 Fa0/1	Access-SW-1 Fa0/1 (uplink)	Straight-through
Dist-SW-1 Fa0/2	Access-SW-2 Fa0/1 (uplink)	Straight-through
Dist-SW-2 Fa0/1	Access-SW-3 Fa0/1 (uplink)	Straight-through
Dist-SW-2 Fa0/2	Access-SW-4 Fa0/1 (uplink)	Straight-through
Access-SW-1 Fa0/2	Admin-PC1	Straight-through
Access-SW-1 Fa0/3	Admin-PC2	Straight-through
Access-SW-2 Fa0/2	IT-PC1	Straight-through
Access-SW-2 Fa0/3	IT-PC2	Straight-through
Access-SW-3 Fa0/2	Lab-PC1	Straight-through
Access-SW-3 Fa0/3	Lab-PC2	Straight-through
Access-SW-4 Fa0/2	Lib-PC1	Straight-through
Dist-SW-2 Fa0/5	TFTP-Server	Straight-through
Dist-SW-2 Fa0/6	DNS/Web Server	Straight-through
Dist-SW-1 Fa0/5	Management-PC	Straight-through

4. End Device IP Configuration

Configure static IP addresses on all PCs and servers. Click on each device > Desktop tab > IP Configuration.

4.1 TFTP Server Configuration

Parameter	Value
IP Address	192.168.100.10
Subnet Mask	255.255.255.0

Default Gateway	192.168.100.1
DNS Server	192.168.100.20

To enable TFTP service: Click Server > Services tab > TFTP > toggle ON.

4.2 DNS / Web Server Configuration

Parameter	Value
IP Address	192.168.100.20
Subnet Mask	255.255.255.0
Default Gateway	192.168.100.1
Services	DNS: ON, HTTP: ON

DNS entry to add: 'university.local' -> 192.168.100.20 (Type A record)

4.3 Management PC

Parameter	Value
IP Address	192.168.50.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.50.1
DNS Server	192.168.100.20

4.4 Department PCs

Device	IP Address	Subnet Mask	Default Gateway
Admin-PC1	192.168.10.10	255.255.255.0	192.168.10.1
Admin-PC2	192.168.10.11	255.255.255.0	192.168.10.1
IT-PC1	192.168.20.10	255.255.255.0	192.168.20.1
IT-PC2	192.168.20.11	255.255.255.0	192.168.20.1
Lab-PC1	192.168.30.10	255.255.255.0	192.168.30.1
Lab-PC2	192.168.30.11	255.255.255.0	192.168.30.1
Lib-PC1	192.168.40.10	255.255.255.0	192.168.40.1

5. Switch Configuration

5.1 Distribution Switch 1 (Dist-SW-1) - L3 Switch

Dist-SW-1 handles VLANs 10 (Admin), 20 (IT), 50 (Management). It routes between these VLANs and has an L3 uplink to the Core Router.

STEP 1 Enter Global Configuration Mode

Dist-SW-1 — Initial Setup & Hostname

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Dist-SW-1
Dist-SW-1(config)# no ip domain-lookup
Dist-SW-1(config)# enable secret cisco123
Dist-SW-1(config)# service password-encryption
Dist-SW-1(config)# banner motd #
=== AUTHORIZED ACCESS ONLY - Dist-SW-1 === #
```

STEP 2 Enable IP Routing on L3 Switch

Dist-SW-1 — Enable Routing

```
Dist-SW-1(config)# ip routing
! Verify routing is enabled
Dist-SW-1(config)# do show ip route
```

STEP 3 Create VLANs

Dist-SW-1 — VLAN Creation

```
Dist-SW-1(config)# vlan 10
Dist-SW-1(config-vlan)# name Admin
Dist-SW-1(config-vlan)# exit
Dist-SW-1(config)# vlan 20
Dist-SW-1(config-vlan)# name IT
Dist-SW-1(config-vlan)# exit
Dist-SW-1(config)# vlan 50
Dist-SW-1(config-vlan)# name Management
Dist-SW-1(config-vlan)# exit
Dist-SW-1(config)# vlan 99
Dist-SW-1(config-vlan)# name Native
Dist-SW-1(config-vlan)# exit
```

STEP 4 Configure SVI (Switch Virtual Interfaces) for Inter-VLAN Routing

Dist-SW-1 — SVI Configuration

```
Dist-SW-1(config)# interface vlan 10
Dist-SW-1(config-if)# description Admin_VLAN_Gateway
Dist-SW-1(config-if)# ip address 192.168.10.1 255.255.255.0
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
Dist-SW-1(config)# interface vlan 20
Dist-SW-1(config-if)# description IT_VLAN_Gateway
Dist-SW-1(config-if)# ip address 192.168.20.1 255.255.255.0
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
Dist-SW-1(config)# interface vlan 50
Dist-SW-1(config-if)# description Management_VLAN_Gateway
Dist-SW-1(config-if)# ip address 192.168.50.1 255.255.255.0
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 5 Configure Routed Uplink Port to Core Router

Dist-SW-1 — Uplink to CR-1

```
Dist-SW-1(config)# interface GigabitEthernet0/1
Dist-SW-1(config-if)# description Uplink_to_CR-1
Dist-SW-1(config-if)# no switchport Dist-SW-1(config-if)# ip
address 10.0.1.2 255.255.255.252 Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 6 Configure Trunk Links to Access Switches

Dist-SW-1 — Trunk Ports (Fa0/1 to Access-SW-1, Fa0/2 to Access-SW-2)

```
Dist-SW-1(config)#interface FastEthernet0/1
Dist-SW-1(config-if)#switchport trunk encapsulation dot1q
Dist-SW-1(config-if)#switchport mode trunk
Dist-SW-1(config-if)#switchport trunk native vlan 99
Dist-SW-1(config-if)#switchport trunk allowed vlan 10,20,50,99
Dist-SW-1(config-if)#no shutdown
Dist-SW-1(config-if)# exit
Dist-SW-1(config)# interface FastEthernet0/2
Dist-SW-1(config-if)# description Trunk_to_Access-SW-2
Dist-SW-1(config-if)# switchport mode trunk
Dist-SW-1(config-if)# switchport trunk native vlan 99
Dist-SW-1(config-if)# switchport trunk allowed vlan 10,20,50,99
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 7 Configure Management PC Access Port

Dist-SW-1 — Management-PC Access Port (Fa0/5)

```
Dist-SW-1(config)# interface FastEthernet0/5
Dist-SW-1(config-if)# description Management-PC
Dist-SW-1(config-if)# switchport mode access
Dist-SW-1(config-if)# switchport access vlan 50
Dist-SW-1(config-if)# spanning-tree portfast
Dist-SW-1(config-if)# no shutdown
Dist-SW-1(config-if)# exit
```

STEP 8 Configure OSPF on Dist-SW-1

Dist-SW-1 — OSPF Configuration

```
Dist-SW-1(config)# router ospf 1
Dist-SW-1(config-router)# router-id 1.1.1.2
Dist-SW-1(config-router)# network 192.168.10.0 0.0.0.255 area 1
Dist-SW-1(config-router)# network 192.168.20.0 0.0.0.255 area 1
Dist-SW-1(config-router)# network 192.168.50.0 0.0.0.255 area 1
Dist-SW-1(config-router)# network 10.0.1.0 0.0.0.3 area 0
Dist-SW-1(config-router)# passive-interface vlan 10
Dist-SW-1(config-router)# passive-interface vlan 20
Dist-SW-1(config-router)# passive-interface vlan 50
Dist-SW-1(config-router)# exit
```

STEP 9 Configure SSH for Remote Management

Dist-SW-1 — SSH Configuration

```
Dist-SW-1(config)# ip domain-name university.local
Dist-SW-1(config)# crypto key generate rsa
How many bits in the modulus [512]: 1024
Dist-SW-1(config)# ip ssh version 2
Dist-SW-1(config)# username admin privilege 15 secret Admin@123
Dist-SW-1(config)# line vty 0 4
Dist-SW-1(config-line)# transport input ssh
Dist-SW-1(config-line)# login local
Dist-SW-1(config-line)# exec-timeout 10 0
Dist-SW-1(config-line)# exit
Dist-SW-1(config)# line console 0
Dist-SW-1(config-line)# password console123
Dist-SW-1(config-line)# login
Dist-SW-1(config-line)# exec-timeout 10 0
Dist-SW-1(config-line)# exit
```

STEP 10 Save Configuration

Dist-SW-1 — Save

```
Dist-SW-1(config)# end
Dist-SW-1# copy running-config startup-config
Destination filename [startup-config]?
[Press Enter]
Building configuration...
[OK]
```

5.2 Distribution Switch 2 (Dist-SW-2) — L3 Switch

Dist-SW-2 handles VLANs 30 (Labs), 40 (Library), 100 (Servers — TFTP, DNS). It connects to Access-SW-3, Access-SW-4, TFTP-Server, and DNS/Web Server.

Dist-SW-2 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Dist-SW-2
Dist-SW-2(config)# no ip domain-lookup
Dist-SW-2(config)# enable secret cisco123
Dist-SW-2(config)# service password-encryption
Dist-SW-2(config)# ip routing !

! --- VLAN Creation ---
Dist-SW-2(config)# vlan 30
Dist-SW-2(config-vlan)# name Labs
Dist-SW-2(config-vlan)# exit
Dist-SW-2(config)# vlan 40
Dist-SW-2(config-vlan)# name Library
Dist-SW-2(config-vlan)# exit
Dist-SW-2(config)# vlan 99
Dist-SW-2(config-vlan)# name Native
Dist-SW-2(config-vlan)# exit
Dist-SW-2(config)# vlan 100
Dist-SW-2(config-vlan)# name Servers
Dist-SW-2(config-vlan)# exit

! --- SVI for Inter-VLAN Routing ---
Dist-SW-2(config)# interface vlan 30
Dist-SW-2(config-if)# description Labs_VLAN_Gateway
Dist-SW-2(config-if)# ip address 192.168.30.1 255.255.255.0
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
Dist-SW-2(config)# interface vlan 40
```

```
Dist-SW-2(config-if)# description Library_VLAN_Gateway
Dist-SW-2(config-if)# ip address 192.168.40.1 255.255.255.0
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
```

```
Dist-SW-2(config)# interface vlan 100
Dist-SW-2(config-if)# description Servers_VLAN_Gateway
Dist-SW-2(config-if)# ip address 192.168.100.1 255.255.255.0
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
```

```
! --- Uplink to CR-1 (L3 Routed Port) ---
Dist-SW-2(config)# interface GigabitEthernet0/1
Dist-SW-2(config-if)# description Uplink_to_CR-1
Dist-SW-2(config-if)# no switchport
Dist-SW-2(config-if)# ip address 10.0.2.2 255.255.255.252
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
```

```
! --- Trunk Ports to Access Switches ---
```

```
Dist-SW-2(config)# interface FastEthernet0/1
Dist-SW-2(config-if)# switchport trunk encapsulation dot1q
Dist-SW-2(config-if)# switchport mode trunk
Dist-SW-2(config-if)# switchport trunk native vlan 99
Dist-SW-2(config-if)# switchport trunk allowed vlan 30,40,99,100
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
```

```
Dist-SW-2(config)# interface FastEthernet0/2
Dist-SW-2(config-if)# switchport trunk encapsulation dot1q
Dist-SW-2(config-if)# switchport mode trunk
Dist-SW-2(config-if)# switchport trunk native vlan 99
Dist-SW-2(config-if)# switchport trunk allowed vlan 30,40,99,100
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
```

```
! --- Server Access Ports ---
Dist-SW-2(config)# interface FastEthernet0/5
Dist-SW-2(config-if)# description TFTP-Server
Dist-SW-2(config-if)# switchport mode access
Dist-SW-2(config-if)# switchport access vlan 100
Dist-SW-2(config-if)# spanning-tree portfast
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
```

```

Dist-SW-2(config)# interface FastEthernet0/6
Dist-SW-2(config-if)# description DNS-Web-Server
Dist-SW-2(config-if)# switchport mode access
Dist-SW-2(config-if)# switchport access vlan 100
Dist-SW-2(config-if)# spanning-tree portfast
Dist-SW-2(config-if)# no shutdown
Dist-SW-2(config-if)# exit
! --- OSPF ---
Dist-SW-2(config)# router ospf 1
Dist-SW-2(config-router)# router-id 1.1.1.3
Dist-SW-2(config-router)# network 192.168.30.0 0.0.0.255 area 2
Dist-SW-2(config-router)# network 192.168.40.0 0.0.0.255 area 2
Dist-SW-2(config-router)# network 192.168.100.0 0.0.0.255 area 2
Dist-SW-2(config-router)# network 10.0.2.0 0.0.0.3 area 0
Dist-SW-2(config-router)# passive-interface vlan 30
Dist-SW-2(config-router)# passive-interface vlan 40
Dist-SW-2(config-router)# passive-interface vlan 100
Dist-SW-2(config-router)# exit

! --- SSH ---
Dist-SW-2(config)# ip domain-name university.local
Dist-SW-2(config)# crypto key generate rsa
How many bits in the modulus [512]: 1024
Dist-SW-2(config)# ip ssh version 2
Dist-SW-2(config)# username admin privilege 15 secret Admin@123
Dist-SW-2(config)# line vty 0 4
Dist-SW-2(config-line)# transport input ssh
Dist-SW-2(config-line)# login local
Dist-SW-2(config-line)# exit

Dist-SW-2(config)# end
Dist-SW-2# write memory

```

5.3 Access Switch 1 (Access-SW-1) — Admin Department

Access-SW-1 — Complete Configuration

```

Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-1
Access-SW-1(config)# no ip domain-lookup
Access-SW-1(config)# enable secret cisco123

```

```
Access-SW-1(config)# service password-encryption
```

```
! --- VLAN Creation ---
```

```
Access-SW-1(config)# vlan 10
Access-SW-1(config-vlan)# name Admin
Access-SW-1(config-vlan)# exit
Access-SW-1(config)# vlan 99
Access-SW-1(config-vlan)# name Native
Access-SW-1(config-vlan)# exit
```

```
! --- Uplink Trunk to Dist-SW-1 ---
```

```
Access-SW-1(config)# interface FastEthernet0/1
Access-SW-1(config-if)# description Trunk_to_Dist-SW-1
Access-SW-1(config-if)# switchport mode trunk
Access-SW-1(config-if)# switchport trunk native vlan 99
Access-SW-1(config-if)# switchport trunk allowed vlan 10,99
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit
```

```
! --- Access Ports for Admin PCs ---
```

```
Access-SW-1(config)# interface FastEthernet0/2
Access-SW-1(config-if)# description Admin-PC1
Access-SW-1(config-if)# switchport mode access
Access-SW-1(config-if)# switchport access vlan 10
Access-SW-1(config-if)# switchport port-security
Access-SW-1(config-if)# switchport port-security maximum 1
Access-SW-1(config-if)# switchport port-security violation
restrict Access-SW-1(config-if)# switchport port-security
mac-address sticky Access-SW-1(config-if)# spanning-tree portfast
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit
```

```
Access-SW-1(config)# interface FastEthernet0/3
Access-SW-1(config-if)# description Admin-PC2
Access-SW-1(config-if)# switchport mode access
Access-SW-1(config-if)# switchport access vlan 10
Access-SW-1(config-if)# switchport port-security
Access-SW-1(config-if)# switchport port-security maximum 1
Access-SW-1(config-if)# switchport port-security violation
restrict Access-SW-1(config-if)# switchport port-security
mac-address sticky Access-SW-1(config-if)# spanning-tree portfast
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit
```

```
! --- Management SVI ---
```

```
Access-SW-1(config)# interface vlan 10
Access-SW-1(config-if)# ip address 192.168.10.250 255.255.255.0
Access-SW-1(config-if)# no shutdown
Access-SW-1(config-if)# exit
Access-SW-1(config)# ip default-gateway 192.168.10.1

! --- Disable unused ports ---
Access-SW-1(config)# interface range FastEthernet0/4-24
Access-SW-1(config-if-range)# shutdown
Access-SW-1(config-if-range)# exit
Access-SW-1(config)# end
Access-SW-1# write memory
```

5.4 Access Switch 2 (Access-SW-2) — IT Department

Access-SW-2 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-2
Access-SW-2(config)# no ip domain-lookup
Access-SW-2(config)# enable secret cisco123
Access-SW-2(config)# service password-encryption

Access-SW-2(config)# vlan 20
Access-SW-2(config-vlan)# name IT
Access-SW-2(config-vlan)# exit
Access-SW-2(config)# vlan 99
Access-SW-2(config-vlan)# name Native
Access-SW-2(config-vlan)# exit

Access-SW-2(config)# interface FastEthernet0/1
Access-SW-2(config-if)# description Trunk_to_Dist-SW-1
Access-SW-2(config-if)# switchport mode trunk
Access-SW-2(config-if)# switchport trunk native vlan 99
Access-SW-2(config-if)# switchport trunk allowed vlan 20,99
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit

Access-SW-2(config)# interface FastEthernet0/2
Access-SW-2(config-if)# description IT-PC1
Access-SW-2(config-if)# switchport mode access
Access-SW-2(config-if)# switchport access vlan 20
Access-SW-2(config-if)# switchport port-security
Access-SW-2(config-if)# switchport port-security maximum 1
```



```

Access-SW-2(config-if)# switchport port-security violation
restrict
Access-SW-2(config-if)# switchport port-security mac-address
sticky
Access-SW-2(config-if)# spanning-tree portfast
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit

Access-SW-2(config)# interface FastEthernet0/3
Access-SW-2(config-if)# description IT-PC2
Access-SW-2(config-if)# switchport mode access
Access-SW-2(config-if)# switchport access vlan 20
Access-SW-2(config-if)# switchport port-security
Access-SW-2(config-if)# switchport port-security maximum 1
Access-SW-2(config-if)# switchport port-security violation
restrict
Access-SW-2(config-if)# switchport port-security mac-address
sticky
Access-SW-2(config-if)# spanning-tree portfast
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit

Access-SW-2(config)# interface vlan 20
Access-SW-2(config-if)# ip address 192.168.20.250 255.255.255.0
Access-SW-2(config-if)# no shutdown
Access-SW-2(config-if)# exit

Access-SW-2(config)# ip default-gateway 192.168.20.1
Access-SW-2(config)# interface range FastEthernet0/4-24
Access-SW-2(config-if-range)# shutdown
Access-SW-2(config-if-range)# exit

Access-SW-2(config)# end
Access-SW-2# write memory

```

5.5 Access Switch 3 (Access-SW-3) — Computer Labs

Access-SW-3 – Complete Configuration

```

Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-3
Access-SW-3(config)# no ip domain-lookup
Access-SW-3(config)# enable secret cisco123

```

```
Access-SW-3(config)# service password-encryption
```

```
Access-SW-3(config)# vlan 30
```

```
Access-SW-3(config-vlan)# name Labs
```

```
Access-SW-3(config-vlan)# exit
```

```
Access-SW-3(config)# vlan 99
```

```
Access-SW-3(config-vlan)# name Native
```

```
Access-SW-3(config-vlan)# exit
```

```
Access-SW-3(config)# interface FastEthernet0/1
```

```
Access-SW-3(config-if)# description Trunk_to_Dist-SW-2
```

```
Access-SW-3(config-if)# switchport mode trunk
```

```
Access-SW-3(config-if)# switchport trunk native vlan 99
```

```
Access-SW-3(config-if)# switchport trunk allowed vlan 30,99
```

```
Access-SW-3(config-if)# no shutdown
```

```
Access-SW-3(config-if)# exit
```

```
Access-SW-3(config)# interface FastEthernet0/2
```

```
Access-SW-3(config-if)# description Lab-PC1
```

```
Access-SW-3(config-if)# switchport mode access
```

```
Access-SW-3(config-if)# switchport access vlan 30
```

```
Access-SW-3(config-if)# spanning-tree portfast
```

```
Access-SW-3(config-if)# no shutdown
```

```
Access-SW-3(config-if)# exit
```

```
Access-SW-3(config)# interface FastEthernet0/3
```

```
Access-SW-3(config-if)# description Lab-PC2
```

```
Access-SW-3(config-if)# switchport mode access
```

```
Access-SW-3(config-if)# switchport access vlan 30
```

```
Access-SW-3(config-if)# spanning-tree portfast
```

```
Access-SW-3(config-if)# no shutdown
```

```
Access-SW-3(config-if)# exit
```

```
Access-SW-3(config)# interface vlan 30
```

```
Access-SW-3(config-if)# ip address 192.168.30.250 255.255.255.0
```

```
Access-SW-3(config-if)# no shutdown
```

```
Access-SW-3(config-if)# exit
```

```
Access-SW-3(config)# ip default-gateway 192.168.30.1
```

```
Access-SW-3(config)# interface range FastEthernet0/4-24
```

```
Access-SW-3(config-if-range)# shutdown
```

```
Access-SW-3(config-if-range)# exit
```

```
Access-SW-3(config)# end
```

```
Access-SW-3# write memory
```

5.6 Access Switch 4 (Access-SW-4) — Library

Access-SW-4 — Complete Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Access-SW-4
Access-SW-4(config)# no ip domain-lookup
Access-SW-4(config)# enable secret cisco123
Access-SW-4(config)# service password-encryption

Access-SW-4(config)# vlan 40
Access-SW-4(config-vlan)# name Library
Access-SW-4(config-vlan)# exit
Access-SW-4(config)# vlan 99
Access-SW-4(config-vlan)# name Native
Access-SW-4(config-vlan)# exit

Access-SW-4(config)# interface FastEthernet0/1
Access-SW-4(config-if)# description Trunk_to_Dist-SW-2
Access-SW-4(config-if)# switchport mode trunk
Access-SW-4(config-if)# switchport trunk native vlan 99
Access-SW-4(config-if)# switchport trunk allowed vlan 40,99
Access-SW-4(config-if)# no shutdown
Access-SW-4(config-if)# exit

Access-SW-4(config)# interface FastEthernet0/2
Access-SW-4(config-if)# description Lib-PC1
Access-SW-4(config-if)# switchport mode access
Access-SW-4(config-if)# switchport access vlan 40
Access-SW-4(config-if)# spanning-tree portfast
Access-SW-4(config-if)# no shutdown
Access-SW-4(config-if)# exit

Access-SW-4(config)# interface vlan 40
Access-SW-4(config-if)# ip address 192.168.40.250 255.255.255.0
Access-SW-4(config-if)# no shutdown
Access-SW-4(config-if)# exit
Access-SW-4(config)# ip default-gateway 192.168.40.1

Access-SW-4(config)# interface range FastEthernet0/3-24
Access-SW-4(config-if-range)# shutdown
Access-SW-4(config-if-range)# exit
Access-SW-4(config)# end
Access-SW-4# write memory
```

6. Router Configuration

6.1 Core Router (CR-1)

CR-1 is the backbone router. It runs OSPF Area 0 (backbone area), connecting to both distribution switches and the edge router via a serial WAN link.

CR-1 — Complete Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname CR-1
CR-1(config)# no ip domain-lookup
CR-1(config)# enable secret cisco123
CR-1(config)# service password-encryption
CR-1(config)# banner motd #
=== AUTHORIZED ACCESS ONLY - Core Router CR-1 === #

! --- Interface to Dist-SW-1 ---
CR-1(config)# interface GigabitEthernet0/0
CR-1(config-if)# description Link_to_Dist-SW-1
CR-1(config-if)# ip address 10.0.1.1 255.255.255.252
CR-1(config-if)# no shutdown
CR-1(config-if)# exit

! --- Interface to Dist-SW-2 ---
CR-1(config)# interface GigabitEthernet0/1
CR-1(config-if)# description Link_to_Dist-SW-2
CR-1(config-if)# ip address 10.0.2.1 255.255.255.252
CR-1(config-if)# no shutdown
CR-1(config-if)# exit

! --- Serial WAN Link to Edge Router ---
CR-1(config)# interface Serial0/0/0
CR-1(config-if)# description WAN_Link_to_ER-1
CR-1(config-if)# ip address 10.0.0.1 255.255.255.252
CR-1(config-if)# clock rate 64000
CR-1(config-if)# no shutdown
CR-1(config-if)# exit

! --- OSPF Configuration ---
CR-1(config)# router ospf 1
CR-1(config-router)# router-id 1.1.1.1
CR-1(config-router)# network 10.0.0.0 0.0.0.3 area 0
CR-1(config-router)# network 10.0.1.0 0.0.0.3 area 0
```

```

CR-1(config-router)# network 10.0.2.0 0.0.0.3 area 0
CR-1(config-router)# exit

! --- SSH Configuration ---
CR-1(config)# ip domain-name university.local
CR-1(config)# crypto key generate rsa
How many bits in the modulus [512]: 1024
CR-1(config)# ip ssh version 2
CR-1(config)# username admin privilege 15 secret Admin@123
CR-1(config)# line vty 0 4
CR-1(config-line)# transport input ssh
CR-1(config-line)# login local
CR-1(config-line)# exec-timeout 10 0
CR-1(config-line)# exit
CR-1(config)# line console 0
CR-1(config-line)# password console123
CR-1(config-line)# login
CR-1(config-line)# exit
CR-1(config)# end
CR-1# write memory

```

6.2 Edge Router (ER-1)

ER-1 acts as the WAN edge device. It connects to CR-1 via serial link and simulates an ISP/Internet connection on its Fa0/0. It also runs OSPF and distributes a default route.

ER-1 — Complete Configuration

```

Router> enable
Router# configure terminal
Router(config)# hostname ER-1
ER-1(config)# no ip domain-lookup
ER-1(config)# enable secret cisco123
ER-1(config)# service password-encryption
ER-1(config)# banner motd #

=== AUTHORIZED ACCESS ONLY - Edge Router ER-1 ===
#

! --- WAN Interface (to ISP/Internet - simulated) ---
ER-1(config)# interface GigabitEthernet0/0
ER-1(config-if)# description Internet_Facing
ER-1(config-if)# ip address 203.0.113.1 255.255.255.0
ER-1(config-if)# ip nat outside

```

```
ER-1(config-if)# no shutdown
ER-1(config-if)# exit

! --- Serial Link to CR-1 ---
ER-1(config)# interface Serial0/0/0
ER-1(config-if)# description WAN_Link_to_CR-1
ER-1(config-if)# ip address 10.0.0.2 255.255.255.252
ER-1(config-if)# ip nat inside
ER-1(config-if)# no shutdown
ER-1(config-if)# exit

! --- Default Route to Internet ---
ER-1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.254

! --- OSPF - Redistribute Default Route ---
ER-1(config)# router ospf 1
ER-1(config-router)# router-id 1.1.1.4
ER-1(config-router)# network 10.0.0.0 0.0.0.3 area 0
ER-1(config-router)# default-information originate
ER-1(config-router)# exit

! --- NAT Configuration ---
! Define which internal traffic gets NATed
ER-1(config)# access-list 10 permit 192.168.0.0 0.0.255.255
ER-1(config)# ip nat inside source list 10 interface
FastEthernet0/0 overload

! --- SSH ---
ER-1(config)# ip domain-name university.local
ER-1(config)# crypto key generate rsa
How many bits in the modulus [512]: 1024
ER-1(config)# ip ssh version 2
ER-1(config)# username admin privilege 15 secret Admin@123
ER-1(config)# line vty 0 4
ER-1(config-line)# transport input ssh
ER-1(config-line)# login local
ER-1(config-line)# exit

ER-1(config)# end
ER-1# write memory
```

7. OSPF Verification & Connectivity Testing

7.1 Verify OSPF Neighbor Relationships

7.1.1 CR-1

User Access Verification

Password:

Password:

CR-1>enable

Password:

CR-1#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.2	1	FULL/DR	00:00:37	10.0.1.2	GigabitEthernet0/0
1.1.1.3	1	FULL/DR	00:00:37	10.0.2.2	GigabitEthernet0/1
1.1.1.4	0	FULL/ -	00:00:37	10.0.0.2	Serial0/0/0

CR-1#show ip route ospf

O IA 192.168.10.0 [110/2] via 10.0.1.2, 00:03:54, GigabitEthernet0/0
O IA 192.168.20.0 [110/2] via 10.0.1.2, 00:03:54, GigabitEthernet0/0
O IA 192.168.30.0 [110/2] via 10.0.2.2, 00:03:54, GigabitEthernet0/1
O IA 192.168.40.0 [110/2] via 10.0.2.2, 00:03:54, GigabitEthernet0/1
O IA 192.168.50.0 [110/2] via 10.0.1.2, 00:03:54, GigabitEthernet0/0
O IA 192.168.100.0 [110/2] via 10.0.2.2, 00:03:54, GigabitEthernet0/1

CR-1#show ip ospf database

OSPF Router with ID (1.1.1.1) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.4	1.1.1.4	279	0x80000003	0x00e8b4	2
1.1.1.1	1.1.1.1	249	0x80000007	0x00bf8b	4
1.1.1.2	1.1.1.2	249	0x80000003	0x0089a4	1
1.1.1.3	1.1.1.3	249	0x80000002	0x009199	1

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.0.1.2	1.1.1.2	249	0x80000001	0x008740
10.0.2.2	1.1.1.3	249	0x80000001	0x008044

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.30.0	1.1.1.3	274	0x80000001	0x006c64
192.168.40.0	1.1.1.3	274	0x80000002	0x00fbc9
192.168.100.0	1.1.1.3	274	0x80000003	0x006325
192.168.10.0	1.1.1.2	280	0x80000001	0x004f96
192.168.20.0	1.1.1.2	280	0x80000002	0x00defb
192.168.50.0	1.1.1.2	280	0x80000003	0x00912a

CR-1#show ip route 192.168.10.0

Routing entry for 192.168.10.0/24

Known via "ospf 1", distance 110, metric 2, type intra area

Last update from 10.0.1.2 on GigabitEthernet0/0, 00:04:57 ago

Routing Descriptor Blocks:

* 10.0.1.2, from 1.1.1.2, 00:04:57 ago, via GigabitEthernet0/0

Route metric is 2, traffic share count is 1

7.1.2 Dist-SW-1

User Access Verification

Password:

Dist-SW-1>enable

Password:

Dist-SW-1#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/BDR	00:00:35	10.0.1.1	GigabitEthernet0/1

Dist-SW-1#show ip route ospf

10.0.0.0/30 is subnetted, 3 subnets

O	10.0.0.0	[110/65]	via 10.0.1.1, 00:07:08,	GigabitEthernet0/1
O	10.0.2.0	[110/2]	via 10.0.1.1, 00:07:08,	GigabitEthernet0/1
O IA	192.168.30.0	[110/3]	via 10.0.1.1, 00:06:58,	GigabitEthernet0/1
O IA	192.168.40.0	[110/3]	via 10.0.1.1, 00:06:58,	GigabitEthernet0/1
O IA	192.168.100.0	[110/3]	via 10.0.1.1, 00:06:58,	GigabitEthernet0/1

Dist-SW-1#show ip ospf database

OSPF Router with ID (1.1.1.2) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.4	1.1.1.4	470	0x80000003	0x00e8b4	2
1.1.1.2	1.1.1.2	440	0x80000003	0x0089a4	1
1.1.1.1	1.1.1.1	440	0x80000007	0x00bf8b	4
1.1.1.3	1.1.1.3	440	0x80000002	0x009199	1

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.0.1.2	1.1.1.2	440	0x80000001	0x008740
10.0.2.2	1.1.1.3	440	0x80000001	0x008044

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.10.0	1.1.1.2	471	0x80000001	0x004f96
192.168.20.0	1.1.1.2	471	0x80000002	0x00defb
192.168.50.0	1.1.1.2	471	0x80000003	0x00912a
192.168.30.0	1.1.1.3	466	0x80000001	0x006c64
192.168.40.0	1.1.1.3	465	0x80000002	0x00fbc9
192.168.100.0	1.1.1.3	465	0x80000003	0x006325

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.2	1.1.1.2	481	0x80000004	0x00d1c5	3

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
10.0.1.0	1.1.1.2	472	0x80000001	0x00cf81
10.0.2.0	1.1.1.2	437	0x80000002	0x00cc81
10.0.0.0	1.1.1.2	437	0x80000004	0x0057b7
192.168.30.0	1.1.1.2	427	0x80000005	0x007e4d
192.168.40.0	1.1.1.2	427	0x80000006	0x000eb2
192.168.100.0	1.1.1.2	427	0x80000007	0x00750e

Summary ASB Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
1.1.1.4	1.1.1.2	437	0x80000003	0x00937b

Dist-SW-1#

Dist-SW-1#show ip route 192.168.10.0

Routing entry for 192.168.10.0/24

Known via "connected", distance 0, metric 0 (connected, via interface)

Routing Descriptor Blocks:

* directly connected, via Vlan10

Route metric is 0, traffic share count is 1

7.1.3 Dist-SW-2

```
Dist-SW-2>enable
Password:
Dist-SW-2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/BDR	00:00:39	10.0.2.1	GigabitEthernet0/1

```
Dist-SW-2#show ip route ospf
10.0.0.0/30 is subnetted, 3 subnets
O    10.0.0.0 [110/65] via 10.0.2.1, 00:13:48, GigabitEthernet0/1
O    10.0.1.0 [110/2] via 10.0.2.1, 00:13:48, GigabitEthernet0/1
O IA 192.168.10.0 [110/3] via 10.0.2.1, 00:13:48, GigabitEthernet0/1
O IA 192.168.20.0 [110/3] via 10.0.2.1, 00:13:48, GigabitEthernet0/1
O IA 192.168.50.0 [110/3] via 10.0.2.1, 00:13:48, GigabitEthernet0/1
```

```
Dist-SW-2#show ip ospf database
OSPF Router with ID (1.1.1.3) (Process ID 1)
```

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.4	1.1.1.4	869	0x80000003	0x00e8b4	2
1.1.1.3	1.1.1.3	839	0x80000002	0x009199	1
1.1.1.1	1.1.1.1	839	0x80000007	0x00bf8b	4
1.1.1.2	1.1.1.2	839	0x80000003	0x0089a4	1

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.0.2.2	1.1.1.3	839	0x80000001	0x008044
10.0.1.2	1.1.1.2	839	0x80000001	0x008740

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.30.0	1.1.1.3	865	0x80000001	0x006c64
192.168.40.0	1.1.1.3	865	0x80000002	0x00fbc9
192.168.100.0	1.1.1.3	865	0x80000003	0x006325
192.168.10.0	1.1.1.2	870	0x80000001	0x004f96
192.168.20.0	1.1.1.2	870	0x80000002	0x00defb
192.168.50.0	1.1.1.2	870	0x80000003	0x00912a

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.30.0	1.1.1.3	865	0x80000001	0x006c64
192.168.40.0	1.1.1.3	865	0x80000002	0x00fbc9
192.168.100.0	1.1.1.3	865	0x80000003	0x006325
192.168.10.0	1.1.1.2	870	0x80000001	0x004f96
192.168.20.0	1.1.1.2	870	0x80000002	0x00defb
192.168.50.0	1.1.1.2	870	0x80000003	0x00912a

Router Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.3	1.1.1.3	880	0x80000005	0x0092a7	3

Summary Net Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
10.0.2.0	1.1.1.3	866	0x80000001	0x00be90
10.0.1.0	1.1.1.3	831	0x80000002	0x00d17c
10.0.0.0	1.1.1.3	831	0x80000004	0x0051bc
192.168.10.0	1.1.1.3	831	0x80000005	0x005589
192.168.20.0	1.1.1.3	831	0x80000006	0x00e4ee
192.168.50.0	1.1.1.3	831	0x80000007	0x00971d

Summary ASB Link States (Area 2)

Link ID	ADV Router	Age	Seq#	Checksum
1.1.1.4	1.1.1.3	831	0x80000003	0x008d80

```
Dist-SW-2#
```

```
Dist-SW-2#show ip route 192.168.10.0
```

```
Routing entry for 192.168.10.0/24
```

```
Known via "ospf 1", distance 110, metric 3, type intra area
```

```
Last update from 10.0.2.1 on GigabitEthernet0/1, 00:14:06 ago
```

```
Routing Descriptor Blocks:
```

```
* 10.0.2.1, from 1.1.1.2, 00:14:06 ago, via GigabitEthernet0/1
```

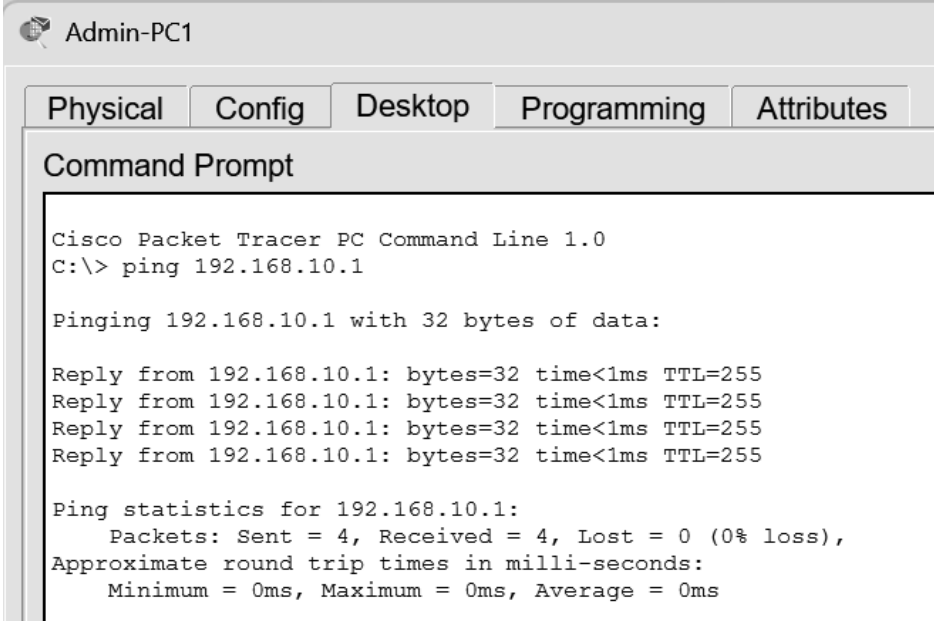
```
Route metric is 3, traffic share count is 1
```

```
Dist-SW-2#
```

7.2 End-to-End Connectivity Tests

Ping Tests from Admin-PC1 (192.168.10.10)

Test 1: Ping default gateway



```
Admin-PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Test 2: Ping another VLAN (IT)

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=36ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 12ms
```

Test 3: Ping TFTP Server

```
C:\>ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.100.10: bytes=32 time=1ms TTL=125
Reply from 192.168.100.10: bytes=32 time=12ms TTL=125
Reply from 192.168.100.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms
```

Test 4: Ping across serial WAN link

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=1ms TTL=253
Reply from 10.0.0.2: bytes=32 time=4ms TTL=253
Reply from 10.0.0.2: bytes=32 time=11ms TTL=253
Reply from 10.0.0.2: bytes=32 time=1ms TTL=253

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms
```

Test 5: Traceroute to verify path

```
C:\>tracert 192.168.30.10

Tracing route to 192.168.30.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.10.1
  1  0 ms    0 ms    0 ms    10.0.1.1
  2  0 ms    12 ms   12 ms   10.0.2.2
  3  *        11 ms   12 ms   192.168.30.10

Trace complete.
```

8. TFTP Configuration Backup

This is the core feature of the project. The TFTP server (192.168.100.10) will store backed-up running and startup configurations from all devices.

8.1 How TFTP Works in CPT

- TFTP (Trivial File Transfer Protocol) uses UDP port 69
- No authentication — relies on network-level security (ACLs)
- In Packet Tracer, the TFTP server must have the TFTP service enabled
- Files are stored on the server and can be retrieved later for restore

8.2 Enable TFTP Service on Server

1. Click on the TFTP-Server in CPT
2. Go to the Services tab
3. Click TFTP on the left panel
4. Toggle the service to ON
5. Note the directory listing — uploaded files will appear here

8.3 CR-1

8.3.1 Backup Running Configuration

```
CR-1# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? CR-1-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

```
CR-1>enable
Password:
CR-1#show flash:

System flash directory:
File   Length   Name/status
  3    33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
  4    33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2     28282   sigdef-category.xml
  1     227537   sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

8.3.2 Backup Startup Configuration

```
CR-1# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? CR-1-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.4 ER-1

8.4.1 Backup Running Configuration

```
ER-1# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? ER-1-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.4.2 Backup Startup Configuration

```
ER-1# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? ER-1-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.5 Dist-SW-1

8.5.1 Backup Running Configuration

```
Dist-SW-1# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? DistSW1-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.5.2 Backup Startup Configuration

```
Dist-SW-1# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? DistSW1-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.6 Dist-SW-2

8.6.1 Backup Running Configuration

```
Dist-SW-2# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? DistSW2-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.6.2 Backup Startup Configuration

```
Dist-SW-2# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? DistSW2-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.7 Access-SW-1

8.7.1 Backup Running Configuration

```
Access-SW-1# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW1-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.7.2 Backup Startup Configuration

```
Access-SW-1# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW1-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.8 Access-SW-2

8.8.1 Backup Running Configuration

```
Access-SW-2# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW2-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.8.2 Backup Startup Configuration

```
Access-SW-2# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW2-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.9 Access-SW-3

8.9.1 Backup Running Configuration

```
Access-SW-3# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW3-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.9.2 Backup Startup Configuration

```
Access-SW-3# copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW3-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.10 Access-SW-4

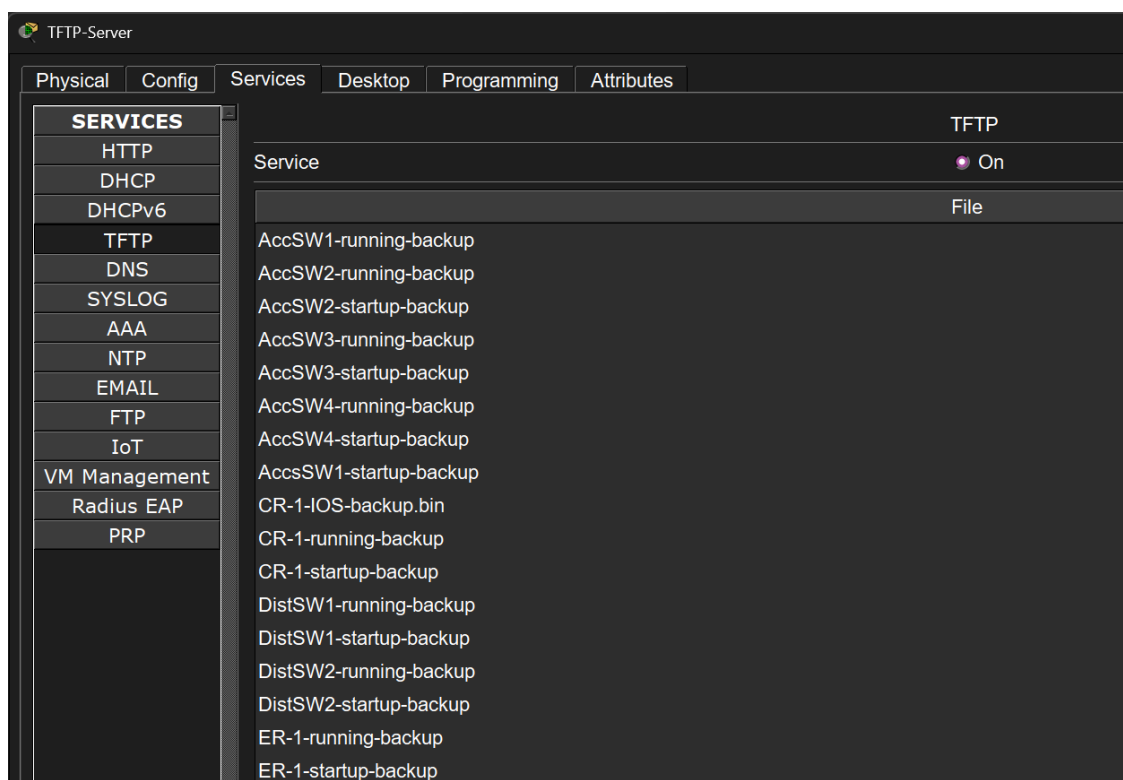
8.10.1 Backup Running Configuration

```
Access-SW-4# copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW4-running-backup
Writing running-config...!!
[OK - 1234 bytes]
```

8.10.2 Backup Startup Configuration

```
Access-SW-4 # copy startup-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [cr-1-config]? AccSW4-startup-backup
Writing startup-config...!!
[OK - 1234 bytes]
```

8.11 Verify Backup on TFTP Server



8.12 Restore Configuration from TFTP

User Access Verification

Password:
Password:

CR-1>enable

Password:

CR-1#copy tftp running-config

Address or name of remote host []? 192.168.100.10

Source filename []? CR-1-running-backup

Destination filename [running-config]?

Accessing tftp://192.168.100.10/CR-1-running-backup...

Loading CR-1-running-backup from 192.168.100.10: !

[OK - 1435 bytes]

1435 bytes copied in 0 secs

CR-1#

%SYS-5-CONFIG_I: Configured from console by console

9. IOS Image Backup and Upgrade via TFTP

9.1 Understanding IOS Image Management

In a production environment, IOS images are large binary files (several MB to hundreds of MB). In Cisco Packet Tracer, IOS images are simulated — the process is demonstrated but actual file sizes and real upgrade behavior may differ from physical hardware.

- IOS image filename format:
[platform]-[feature_set]-[format].[version].bin
- Example: c2900-universalk9-mz.SPA.151-4.M4.bin
- TFTP is the traditional method for IOS transfers in enterprise environments
- In CPT, the 'show version' command displays the simulated IOS version

9.2 Check Current IOS Version

CR-1 — Verify Current IOS Version

```
CR-1# show version Cisco IOS Software, Version 15.1(4)M4, RELEASE  
SOFTWARE (fc1)
```

```
Technical Support: http://www.cisco.com/techsupport
```

```
...
```

```
ROM: System Bootstrap, Version 15.0(1)M3
```

```
...
```

```
System image file is
```



```
CR-1#copy tftp flash:  
Address or name of remote host []? 192.168.100.10  
Source filename []? c2900-universalk9-mz.SPA.155-3.M.bin  
Destination filename [c2900-universalk9-mz.SPA.155-3.M.bin]?  
  
Accessing tftp://192.168.100.10/c2900-universalk9-mz.SPA.155-3.M.bin...  
%Error opening tftp://192.168.100.10/c2900-universalk9-mz.SPA.155-3.M.bin (No such file or directory)  
CR-1#copy tftp flash:  
Address or name of remote host []? 192.168.100.10  
Source filename []? c2900-universalk9-mz.SPA.155-3.M4a.bin  
Destination filename [c2900-universalk9-mz.SPA.155-3.M4a.bin]?  
  
Accessing tftp://192.168.100.10/c2900-universalk9-mz.SPA.155-3.M4a.bin...  
Loading c2900-universalk9-mz.SPA.155-3.M4a.bin from 192.168.100.10:  
|!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!|  
|!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!|  
|!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!|  
|!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!|  
|!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!|  
|!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!|  
[OK - 33591768 bytes]  
  
33591768 bytes copied in 1.48 secs (2383106 bytes/sec)  
CR-1#
```

9.5 Configure Boot System for New IOS

After copying the new image, configure the router to boot from it on the next reload.

CR-1 — Set Boot System & Reload

```
CR-1# configure terminal
! Set boot order: try new image first, fall back to old
CR-1(config)# boot system
flash:c2900-universalk9-mz.SPA.155-3.M.bin
CR-1(config)# boot system
flash:c2900-universalk9-mz.SPA.151-4.M4.bin CR-1(config)# end
```

```
CR-1#show running-config | include boot
boot system flash c2900-universalk9-mz.SPA.155-3.M4a.bin
boot system flash c2900-universalk9-mz.SPA.151-4.M4.bin
CR-1#
```

10. Advanced Configuration & Security

10.1 Access Control Lists (ACLs)

ACLs restrict which devices can access the TFTP server and management interfaces, improving security.

CR-1 — ACL: Restrict TFTP Access to Management VLAN Only

```
CR-1(config)# ip access-list extended TFTP-ACCESS
CR-1(config-ext-nacl)# remark -- Allow Management VLAN to TFTP
server --
CR-1(config-ext-nacl)# permit udp 192.168.50.0 0.0.0.255 host
192.168.100.10 eq 69
CR-1(config-ext-nacl)# remark -- Allow all internal VLANs to reach
TFTP --
CR-1(config-ext-nacl)# permit udp 192.168.0.0 0.0.255.255 host
192.168.100.10
eq 69
CR-1(config-ext-nacl)# deny ip any host 192.168.100.10 log
CR-1(config-ext-nacl)# permit ip any any
CR-1(config-ext-nacl)# exit
```

10.2 Spanning Tree Protocol (STP) Optimization

Dist-SW-1 — STP Root Bridge Configuration

```
! Make Dist-SW-1 the root bridge for VLANs 10, 20, 50
Dist-SW-1(config)# spanning-tree vlan 10 root primary
Dist-SW-1(config)# spanning-tree vlan 20 root primary
Dist-SW-1(config)# spanning-tree vlan 50 root primary

! Make Dist-SW-2 the root bridge for VLANs 30, 40, 100
! (run on Dist-SW-2)
Dist-SW-2(config)# spanning-tree vlan 30 root primary
Dist-SW-2(config)# spanning-tree vlan 40 root primary
Dist-SW-2(config)# spanning-tree vlan 100 root primary

! Enable BPDU Guard on all access ports
Dist-SW-1(config)# spanning-tree portfast bpduguard default
```

10.3 DHCP Configuration on ER-1

Optionally, configure DHCP pools on ER-1 or Dist-SW-1 for dynamic IP assignment to end devices.

Dist-SW-1 — DHCP Pool for Admin VLAN

```
Dist-SW-1(config)# ip dhcp pool ADMIN-POOL
Dist-SW-1(dhcp-config)# network 192.168.10.0 255.255.255.0
Dist-SW-1(dhcp-config)# default-router 192.168.10.1
Dist-SW-1(dhcp-config)# dns-server 192.168.100.20
Dist-SW-1(dhcp-config)# lease 7
Dist-SW-1(dhcp-config)# exit

! Exclude static IPs from DHCP range
Dist-SW-1(config)# ip dhcp excluded-address 192.168.10.1
192.168.10.20
Dist-SW-1(config)# ip dhcp pool IT-POOL
Dist-SW-1(dhcp-config)# network 192.168.20.0 255.255.255.0
Dist-SW-1(dhcp-config)# default-router 192.168.20.1
Dist-SW-1(dhcp-config)# dns-server 192.168.100.20
Dist-SW-1(dhcp-config)# lease 7
Dist-SW-1(dhcp-config)# exit
Dist-SW-1(config)# ip dhcp excluded-address 192.168.20.1
192.168.20.20
```

11. Complete Verification

11.1 Switch Verification

11.1.1 Verify VLANs

```
Dist-SW-1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	Admin	active	
20	IT	active	
50	Management	active	Fa0/5
99	Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

11.1.2 Verify trunk links

```
Dist-SW-1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    99
Fa0/2     on        802.1q         trunking    99

Port      Vlans allowed on trunk
Fa0/1     10,20,50,99
Fa0/2     10,20,50,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,50,99
Fa0/2     10,20,50,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,50,99
Fa0/2     10,20,50,99
```

11.1.3 Verify SVI status

```
Dist-SW-1#show interfaces vlan 10
Vlan10 is up, line protocol is up
Hardware is CPU Interface, address is 0030.f267.ba01 (bia 0030.f267.ba01)
Description: Admin_VLAN_Gateway
Internet address is 192.168.10.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
 1682 packets input, 530955 bytes, 0 no buffer
   Received 0 broadcasts (0 IP multicast)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
 563859 packets output, 0 bytes, 0 underruns
    0 output errors, 23 interface resets
    0 output buffer failures, 0 output buffers swapped out
```

```

Dist-SW-1#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/1          unassigned      YES unset  up            up
FastEthernet0/2          unassigned      YES unset  up            up
FastEthernet0/3          unassigned      YES unset  down          down
FastEthernet0/4          unassigned      YES unset  down          down
FastEthernet0/5          unassigned      YES unset  up            up
FastEthernet0/6          unassigned      YES unset  down          down
FastEthernet0/7          unassigned      YES unset  down          down
FastEthernet0/8          unassigned      YES unset  down          down
FastEthernet0/9          unassigned      YES unset  down          down
FastEthernet0/10         unassigned      YES unset  down          down
FastEthernet0/11         unassigned      YES unset  down          down
FastEthernet0/12         unassigned      YES unset  down          down
FastEthernet0/13         unassigned      YES unset  down          down
FastEthernet0/14         unassigned      YES unset  down          down
FastEthernet0/15         unassigned      YES unset  down          down
FastEthernet0/16         unassigned      YES unset  down          down
FastEthernet0/17         unassigned      YES unset  down          down
FastEthernet0/18         unassigned      YES unset  down          down
FastEthernet0/19         unassigned      YES unset  down          down
FastEthernet0/20         unassigned      YES unset  down          down
FastEthernet0/21         unassigned      YES unset  down          down
FastEthernet0/22         unassigned      YES unset  down          down
FastEthernet0/23         unassigned      YES unset  down          down
FastEthernet0/24         unassigned      YES unset  down          down
GigabitEthernet0/1       10.0.1.2        YES manual up            up
GigabitEthernet0/2       unassigned      YES unset  down          down
Vlan1                    unassigned      YES unset  administratively down down
Vlan10                   192.168.10.1    YES manual up            up
Vlan20                   192.168.20.1    YES manual up            up
Vlan50                   192.168.50.1    YES manual up            up

```

11.1.4 Verify port security

```

Access-SW-1>enable
Password:
Access-SW-1#show port-security interface FastEthernet0/2
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 000D.BD14.0DE5:10
Security Violation Count : 0

```

```

Access-SW-1#show port-security address
Secure Mac Address Table

```

Vlan	Mac Address	Type	Ports	Remaining Age (mins)
10	000D.BD14.0DE5	SecureSticky	Fa0/2	-

```

Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024

```

11.1.5 Verify spanning tree

```
Dist-SW-1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
    Root ID    Priority    24586
              Address     0030.F267.BAC9
              This bridge is the root
              Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec

    Bridge ID  Priority    24586 (priority 24576 sys-id-ext 10)
              Address     0030.F267.BAC9
              Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
              Aging Time  20

Interface                Role Sts Cost          Prio.Nbr Type
-----
Fa0/1                    Desg FWD 19           128.1   P2p
Fa0/2                    Desg FWD 19           128.2   P2p
```

11.1.6 Verify CDP neighbors

```
Dist-SW-1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID    Local Intrfce    Holdtme    Capability    Platform    Port ID
Access-SW-2  Fas 0/2          165        S             2960        Fas 0/1
CR-1         Gig 0/1          166        R             C2900        Gig 0/0
Access-SW-1  Fas 0/1          165        S             2960        Fas 0/1

Dist-SW-1#show cdp neighbors detail

Device ID: Access-SW-2
Entry address(es):
  IP address : 192.168.20.250
Platform: cisco 2960, Capabilities: Switch
Interface: FastEthernet0/2, Port ID (outgoing port): FastEthernet0/1
Holdtime: 136

Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

advertisement version: 2
Duplex: full
-----

Device ID: CR-1
Entry address(es):
  IP address : 10.0.1.1
Platform: cisco C2900, Capabilities: Router
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0
Holdtime: 136

Version :
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE (fc1))
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Thu 06-Oct-16 14:43 by mnguyen

advertisement version: 2
Duplex: full
-----

Device ID: Access-SW-1
Entry address(es):
  IP address : 192.168.10.250
Platform: cisco 2960, Capabilities: Switch
Interface: FastEthernet0/1, Port ID (outgoing port): FastEthernet0/1
Holdtime: 136

Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

advertisement version: 2
Duplex: full
```

11.2 Router Verification Router

11.2.1 Verify interface status

```
CR-1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.0.1.1        YES manual up          up
GigabitEthernet0/1       10.0.2.1        YES manual up          up
GigabitEthernet0/2       unassigned      YES NVRAM  administratively down down
Serial0/0/0              10.0.0.1        YES manual up          up
Serial0/0/1              unassigned      YES NVRAM  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
```

11.2.2 Verify OSPF

```
CR-1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.2	1	FULL/DR	00:00:32	10.0.1.2	GigabitEthernet0/0
1.1.1.3	1	FULL/DR	00:00:32	10.0.2.2	GigabitEthernet0/1
1.1.1.4	0	FULL/-	00:00:35	10.0.0.2	Serial0/0/0

```
CR-1#show ip ospf
```

```
Routing Process "ospf 1" with ID 1.1.1.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm executed 24 times
    Area ranges are
    Number of LSA 12. Checksum Sum 0x066984
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
```

```
CR-1#show ip route ospf
```

```
O IA 192.168.10.0 [110/2] via 10.0.1.2, 02:46:47, GigabitEthernet0/0
O IA 192.168.20.0 [110/2] via 10.0.1.2, 02:46:47, GigabitEthernet0/0
O IA 192.168.30.0 [110/2] via 10.0.2.2, 02:46:47, GigabitEthernet0/1
O IA 192.168.40.0 [110/2] via 10.0.2.2, 02:46:47, GigabitEthernet0/1
O IA 192.168.50.0 [110/2] via 10.0.1.2, 02:46:47, GigabitEthernet0/0
O IA 192.168.100.0 [110/2] via 10.0.2.2, 02:46:47, GigabitEthernet0/1
```


11.2.3 Verify routing table

```
CR-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet0/0
L       10.0.1.1/32 is directly connected, GigabitEthernet0/0
C       10.0.2.0/30 is directly connected, GigabitEthernet0/1
L       10.0.2.1/32 is directly connected, GigabitEthernet0/1
O IA 192.168.10.0/24 [110/2] via 10.0.1.2, 02:47:32, GigabitEthernet0/0
O IA 192.168.20.0/24 [110/2] via 10.0.1.2, 02:47:32, GigabitEthernet0/0
O IA 192.168.30.0/24 [110/2] via 10.0.2.2, 02:47:32, GigabitEthernet0/1
O IA 192.168.40.0/24 [110/2] via 10.0.2.2, 02:47:32, GigabitEthernet0/1
O IA 192.168.50.0/24 [110/2] via 10.0.1.2, 02:47:32, GigabitEthernet0/0
O IA 192.168.100.0/24 [110/2] via 10.0.2.2, 02:47:32, GigabitEthernet0/1
```

11.2.4 Verify SSH sessions

```
CR-1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
```

11.2.5 Verify NAT translations (on ER-1)

```
ER-1>enable
Password:
ER-1#show ip nat translations
ER-1#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/0
Inside Interfaces: Serial0/0/0
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
```


11.2.6 Verify ACLs

```
CR-1#show access-lists
Extended IP access list TFTP-ACCESS
 10 permit udp 192.168.50.0 0.0.0.255 host 192.168.100.10 eq tftp
 20 permit udp 192.168.0.0 0.0.255.255 host 192.168.100.10 eq tftp
 30 permit ip any any
 40 deny ip any host 192.168.100.10

CR-1#show ip access-lists TFTP-ACCESS
Extended IP access list TFTP-ACCESS
 permit udp 192.168.50.0 0.0.0.255 host 192.168.100.10 eq tftp
 permit udp 192.168.0.0 0.0.255.255 host 192.168.100.10 eq tftp
 permit ip any any
 deny ip any host 192.168.100.10
```

11.3 TFTP Verification

11.3.1 List files in flash

```
CR-1#show flash:

System flash directory:
File   Length   Name/status
  3    33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
  4    33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
  2     28282   sigdef-category.xml
  1     227537   sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

11.3.2 Verify config was saved

```
CR-1#show startup-config
Using 1550 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname CR-1
!
!
boot system flash c2900-universalk9-mz.SPA.155-3.M4a.bin
boot system flash c2900-universalk9-mz.SPA.151-4.M4.bin
!
```

```
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username admin privilege 15 secret 5
$1$mERr$kku/hiJPh.4Yp8Xor/4kz1
!
!
license udi pid CISC02911/K9 sn FTX1524AC1K-
!
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
no ip domain-lookup
ip domain-name university.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
description Link_to_Dist-SW-1
ip address 10.0.1.1 255.255.255.252
duplex auto
speed auto
```

```
!  
interface GigabitEthernet0/1  
description Link_to_Dist-SW-2  
ip address 10.0.2.1 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Serial0/0/0  
description WAN_Link_to_ER-1  
ip address 10.0.0.1 255.255.255.252  
clock rate 64000  
!  
interface Serial0/0/1  
no ip address  
clock rate 2000000  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
router-id 1.1.1.1  
log-adjacency-changes  
network 10.0.0.0 0.0.0.3 area 0  
network 10.0.1.0 0.0.0.3 area 0  
network 10.0.2.0 0.0.0.3 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
!  
!  
banner motd ^C  
=== AUTHORIZED ACCESS ONLY - Core Router CR-1 ===  
^C  
!  
!
```

```
!  
!  
line con 0  
password 7 082243401A16091243595F  
login  
!  
line aux 0  
!  
line vty 0 4  
login local  
transport input ssh  
!  
!  
!  
end
```

11.3.3 Verify running config

```
CR-1#show running-config  
Building configuration...  
Current configuration : 1988 bytes  
!  
version 15.3  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
service password-encryption  
!  
hostname CR-1  
!  
!  
boot system flash c2900-universalk9-mz.SPA.155-3.M4a.bin  
boot system flash c2900-universalk9-mz.SPA.151-4.M4.bin  
boot system flash c2900-universalk9-mz.SPA.155-3.M.bin  
!  
enable secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/  
!  
!  
!  
!  
!  
!  
no ip cef  
no ipv6 cef  
!
```

```
!  
!  
username admin privilege 15 secret 5  
$1$mERr$kkku/hiJPh.4Yp8Xor/4kz1  
!  
!  
license udi pid CISC02911/K9 sn FTX1524AC1K-  
license boot module c2900 technology-package FoundationSuiteK9  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip ssh version 2  
no ip domain-lookup  
ip domain-name university.local  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
description Link_to_Dist-SW-1  
ip address 10.0.1.1 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
description Link_to_Dist-SW-2  
ip address 10.0.2.1 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address
```

```
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
description WAN_Link_to_ER-1
ip address 10.0.0.1 255.255.255.252
clock rate 64000
!
interface Serial0/0/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 1.1.1.1
log-adjacency-changes
network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.3 area 0
network 10.0.2.0 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended TFTP-ACCESS
remark -- Allow Management VLAN to TFTP server --
permit udp 192.168.50.0 0.0.0.255 host 192.168.100.10 eq tftp
remark -- Allow all internal VLANs to reach TFTP --
permit udp 192.168.0.0 0.0.255.255 host 192.168.100.10 eq tftp
permit ip any any
deny ip any host 192.168.100.10
!
banner motd ^C
=== AUTHORIZED ACCESS ONLY - Core Router CR-1 ===
^C
!
!
!
!
```

```
line con 0
password 7 082243401A16091243595F
login
!
line aux 0
!
line vty 0 4
login local
transport input ssh
!
!
!
end
```

12. Simulation Mode in Packet Tracer

12.1 Setting Up Simulation

1. Click the Simulation tab in the bottom-right corner of CPT (looks like a stopwatch icon)
2. In the Event List Filters, select the protocols you want to trace: ICMP, TFTP, OSPF
3. From a PC, initiate a ping or TFTP operation
4. Click Play or Step to advance the simulation
5. Watch packets flow through the topology with colored envelopes

12.2 Simulating TFTP Backup

1. Switch to Simulation Mode
2. Add filter: TFTP (UDP/69)
3. Open CLI on CR-1 and type: copy running-config tftp
4. Enter TFTP server IP and filename
5. Click Play in Simulation panel
6. Observe TFTP packets traveling from CR-1 through the network to the TFTP Server
7. Click on any packet envelope to inspect PDU details

12.3 Simulating OSPF Convergence

1. Switch to Simulation Mode
2. Add filter: OSPF 20. Power cycle (shutdown/no shutdown) an interface on CR-1
3. Click Play to watch OSPF hello packets, LSAs, and reconvergence

12.4 Verifying Inter-VLAN Traffic Path

1. Switch to Simulation Mode
2. Add filter: ICMP
3. From Admin-PC1, ping Library-PC1 (192.168.40.10)
4. Observe: PC1 > Access-SW-1 > Dist-SW-1 > CR-1 > Dist-SW-2 > Access-SW-4 > Lib PC1
5. Click on each packet to verify the IP headers and encapsulation at each hop